Eliot Kim

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EDUCATION

University of Washington **Sep 2024 – Present**
Ph.D., Atmospheric and Climate Science *Seattle, WA*

University of Washington **Sep. 2024 – May 2026**
M.S., Atmospheric and Climate Science *Seattle, WA*

University of Wisconsin-Madison **Sep. 2019 – May 2023**
B.S., Computer Science; B.S., Statistics; B.S., Mathematics *Madison, WI*

RESEARCH INTERESTS

Adapting the latest developments in data-driven modeling to understand Earth system processes and predict future climates.

RESEARCH EXPERIENCES

Graduate Research Assistant **September 2024 — Present**
Advisor: Prof. Alexander Turner *University of Washington*

- Fine-tuning an Earth System foundation model using output from chemistry-climate models to constrain oxidative capacity and understand recent variability in methane trends.

Research Assistant **September 2023 — August 2024**
Advisors: Drs. Christoph Keller, Julie Nicely *Global Modeling and Assimilation Office, NASA GSFC*

- Adapted machine learning methods to emulate the GEOS-Chem chemical mechanism and optimize satellite data assimilation.
- Used a chemical box model to generate data and investigate chemical relationships.
- Investigated the ability of machine learning models to capture chemical relationships.

Research Assistant **May 2020 – August 2023**
Advisor: Prof. Tracey Holloway *University of Wisconsin-Madison*

- Investigated the ability of satellite NO_2 data supplemented with location-specific information to estimate the concentration and distribution of NO_2 at the surface.
- Developed machine learning models to classify intensity of emissions plumes over large Texas coal and natural gas power plants using satellite images.
- Updated publicly available WHIPS (Wisconsin Horizontal Interpolation for Satellites) Python package to re-grid several satellite air pollutant products onto custom grids. Enables comparison between satellite and model simulated data.
- Analyzed satellite, model, and emissions inventory NOx and SOx data over Texas power plants to characterize ability of satellites to capture point source pollution.

Research Intern **May 2022 – February 2023**
Advisors: Drs. David John Gagne and John Schreck *National Center for Atmospheric Research*

- Interned in Analytics and Integrative Machine Learning group mentored by Drs. John Schreck and David John Gagne.
- Implemented neural networks with evidential loss functions to predict precipitation type and amount of evidence for each prediction. Analyzed atmospheric sounding inputs from the Rapid Refresh model.
- Used Explainable AI methods (SHAP, Permutation Importance) to analyze the physical consistency of neural network predictions and found physical structure in feature attributions.
- Implemented Active Learning to improve neural network training efficiency by training on the training examples with greatest uncertainty.
- Achieved three-fold accuracy improvement for freezing rain prediction on 20% of the full dataset.

Research Intern **June 2021– September 2021**
Advisor: Prof. Jianwu Wang *NSF Big Data REU, University of Maryland, Baltimore County*

- Harnessed deep learning methods (CNN, ConvLSTM) to predict spatiotemporal Arctic sea ice trends based on ERA5 data. Deployed and optimized models using slurm, Horovod, and Apache Spark.
- Developed multi-task deep learning models and custom loss functions and achieved accuracy improvements over prior literature.

PUBLICATIONS

* = equal contribution

1. **Eliot Kim**, Alexander Turner, TODO (in prep). Emulating Chemistry and Climate with an Isotope-Enabled Foundation Model.

2. **Eliot Kim**; Christoph Keller; Julie Nicely; Peter Ivatt; Michael Long; Obin Sturm; Melanie Follette-Cook; Steven Pawson (under review). Machine Learning-Based Operator for Chemical Data Assimilation in the NASA GEOS Modeling System.

3. Charlie Becker, John S. Schreck, David John Gagne II, Gabrielle Gantos, **Eliot Kim**, Dhamma Kimpara, Jacob Radford, Belen Saavedra, Justin Willson, Christopher D. Wirz, and Julie L. Demuth (accepted). Winter Precipitation Type Diagnosis and Uncertainty Quantification with a Physically Consistent Machine Learning Method.

4. John Schreck; David John Gagne; Charlie Becker; William Chapman; Kim Elmore; Da Fan; Gabrielle Gantos; **Eliot Kim**; Dhamma Kimpara; Thomas Martin; Maria Molina; Vanessa Przybylo; Jacob Radford; Belen Saavedra; Justin Willson; Christopher Wirz (2024). Evidential Deep Learning: Enhancing Predictive Uncertainty Estimation for Earth System Science Applications. *Artificial Intelligence for the Earth Systems.* 10.1175/AIES-D-23-0093.1.

5. **Eliot Kim**; Ajinkya Kokandakar; Tracey Holloway; Steph Elkins; Monica Harkey; Daniel Goldberg; Colleen Heck (2024). A Comparison of Regression Methods for Inferring Near-Surface NO_2 with Satellite Data. *Journal of Geophysical Research: Atmospheres.* 129. 10.1029/2024JD040906

6. Mitch Bushuk et al. [61 co-authors] (2024). Predicting September Arctic Sea Ice: A Multi-Model Seasonal Skill Comparison. *Bulletin of the American Meteorological Society.*

7. ***Eliot Kim**; *Peter Kruse; Skylar Lama; Jamal Bourne; Michael Hu; Sahara Ali; Yiyi Huang; Jianwu Wang (2021). Multi-Task Spatiotemporal Deep Learning Based Arctic Sea Ice Prediction. *2021 IEEE International Conference on Big Data (Big Data).* 1847. 10.1109/BigData52589.2021.9671491.

8. Monica Harkey; Tracey Holloway; **Eliot Kim**; Kirk Baker; Barron Henderson (2020). Satellite Formaldehyde to Support Model Evaluation. *Journal of Geophysical Research: Atmospheres.* 126. 10.1029/2020JD032881.

PRESENTATIONS

1. Enabling joint chemical and meteorological data assimilation with an atmospheric foundation model. Invited Talk. *JPL Carbon Club*, Pasadena, CA, Nov 2026.

2. Emulating Chemistry and Climate with an Isotope-Enabled Foundation Model. Invited Talk. *Machine Learning for Atmospheric Chemistry Workshop*, Reading, UK, Jul 2026.

3. Emulating Chemistry and Climate with an Isotope-Enabled Foundation Model. Invited Talk. *Allen Institute for Artificial Intelligence*, Seattle, WA, Jun 2026.

4. Emulating Chemistry and Climate with an Isotope-Enabled Foundation Model. Oral Presentation. *12th International GEOS-Chem Meeting*, St. Louis, MO, Jun 2026.

5. Emulating Chemistry and Climate with an Isotope-Enabled Foundation Model. Oral Presentation. *Atmospheric Physics and Chemistry Seminar, UW Department of Atmospheric and Climate Science*, Seattle, WA, Apr 2026.

6. Emulating Chemistry and Climate with an Isotope-Enabled Foundation Model. Oral Presentation. *FETCH4 Mid-Term Review*, Rochester, NY, Apr 2026.

7. Fine-tuning an Earth System foundation model to constrain the atmospheric oxidative capacity. Oral Presentation. *FETCH4 Seminar*, Virtual, Mar 2026.

8. Fine-tuning an Earth System foundation model to constrain the atmospheric oxidative capacity. Poster. *American Geophysical Union Annual Meeting 2025*, New Orleans, LA, Dec 2025.

9. Machine Learning-Based Adjoint Emulation for Chemical Data Assimilation in the NASA GEOS Modeling System. Poster. *American Geophysical Union Annual Meeting 2024*, Washington, D.C., Dec 2024.

10. Machine Learning-Based Adjoint Emulation for Chemical Data Assimilation in the NASA GEOS Modeling System. Oral Presentation. *Statistical Learning in Atmospheric Chemistry Group*, Virtual, Sep 2024.

11. Machine Learning-Based Operator for Chemical Data Assimilation in the NASA GEOS Modeling System. Oral Presentation. *NASA-UIUC Machine Learning Applications for Air Quality Workshop*, Virtual, Feb 2024.

12. First Steps Toward Implementing a Machine Learning-Based Operator for Chemical Data Assimilation in the NASA GEOS Modeling System. Oral Presentation. *American Geophysical Union Annual Meeting 2023*, San Francisco, CA, Dec 2023.

13. Explainable and Efficient Neural Networks for Predicting Winter Weather Precipitation Type. Poster. *American Meteorological Society 103rd Annual Meeting*, Denver, CO, Jan 2023.

AWARDS

College of Environment Student Travel Award **June 2026**
University of Washington
- Awarded to present at 12th International GEOS-Chem Conference.

Graduate Student Distinguished Service Award **September 2025**
University of Washington
- Awarded for service on Welcome Committee for incoming first-year cohort.

College of Environment Endowed Scholarship **September 2024**
University of Washington
- Awarded top-off funding as one of the top applicants to the College of Environment graduate programs for Fall 2024.

Hilldale Research Fellowship **August 2022**
University of Wisconsin-Madison
- Awarded research and travel funding for project to study agreement between TROPOMI satellite data and surface-level NO_2 pollution.

IEEE BigData 2021 Student Travel Award **December 2021**
University of Maryland, Baltimore County
- Support for attending and presenting at the 2021 IEEE International Conference on Big Data.

Welton Summer Sophomore Apprenticeship **May 2020**
University of Wisconsin-Madison
- Awarded for project to compare Aerosol Optical Depth from satellite observations and air quality models.

RESEARCH COMPETITIONS

ProjectX **September 2020 – January 2021; September 2021 – January 2022**

- International machine learning research competition for undergraduates organized annually by University of Toronto AI Club. Co-led a team of five other UW-Madison students, Dr. Steven Schrodi, and Dr. Mark Keller.
- **2021**: Trained artificial neural networks to predict glucose levels in mice based on gene expression data across five tissues. Conducted feature importance analysis to determine the metabolic pathways associated with glucose levels.
- **2020**: Implemented machine learning models (Random Forests, ANN, Convolutional LSTM) to predict spread of wildfire air pollutants in Northern California based on atmospheric and meteorological variables.

LEADERSHIP

UW Atmospheric and Climate Science Outreach Team **Fall 2025 – Present**

Coordinator *University of Washington*

- Organizing and hosting educational field trips for K-12 and undergraduate students.

Undergraduate Statistics Club **Spring 2020 – Fall 2022**

Vice President and President *University of Wisconsin-Madison*

- Organized speaker events, student panels, workshops, and month-long data analysis competition.

Data Science Club **Spring 2020 – Fall 2022**

Board Member and Co-President *University of Wisconsin-Madison*

- Inaugurated a data science hackathon, organized student project opportunities, and hosted industry speakers and workshops.

RELEVANT COURSEWORK

Statistics

- Deep Learning; Machine Learning; Matrix Methods for Machine Learning; Time Series Analysis; Data-Driven Dynamical Systems

Computer Science

- High-Performance Comuting; Operating Systems; Algorithms; Database Management; Data Structures

Mathematics

- Combinatorics; Real Analysis; Topology; Stochastic Processes; Probability Theory; Linear Optimization; Scientific Computing

Earth Science

- Air Quality Analysis; Atmospheric and Oceanic Dynamics; Atmospheric Radiation; Atmospheric Chemistry; Cloud Physics; Carbon Cycle; Synoptic Meteorology

PROFESSIONAL MEMBERSHIPS

American Geophysical Union **2022 - Present**

TOOLS AND SKILLS

Programming: Python, C, Java, JavaScript, R, Fortran, SQL, bash
Computing: Git/GitHub, slurm, PBS, Apache Spark, Docker, xarray
Modeling: PyTorch, TensorFlow, scikit-learn
Languages: English (native), Korean (fluent), Spanish (intermediate)